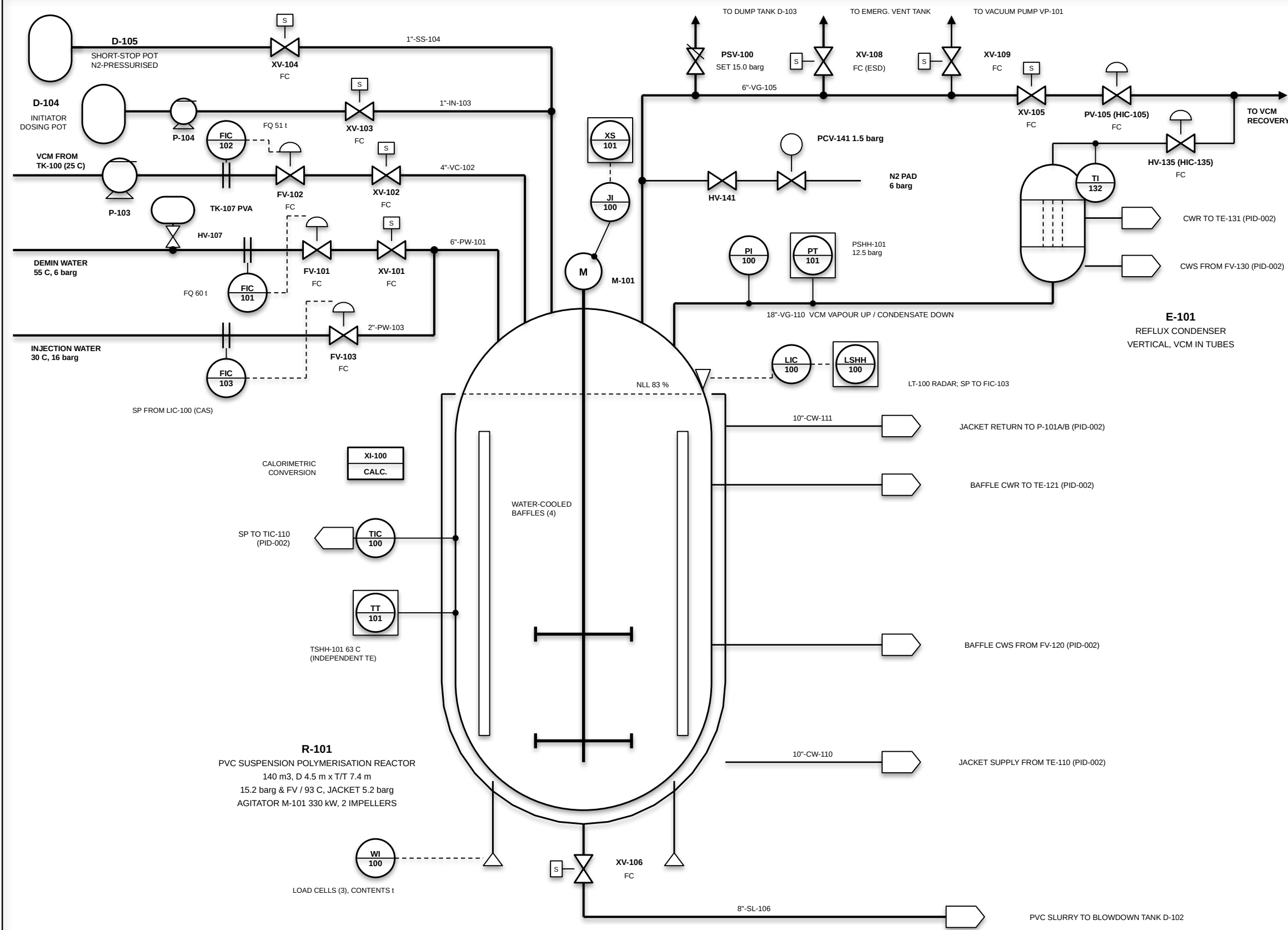




GENERAL NOTES:

1. DESIGN BASIS (REFERENCE CARD): VINYL CHLORIDE IN WATER -> PVC; JACKETED BATCH TANK + REFLUX CONDENSER; 50-70 C, 8-12 bar; ORGANIC PEROXIDE INITIATOR; 200 kJ/y -> 7.9 MW (dHp 71 kJ/mol, 8000 h/y). PLANT = 4 x 140 m3 REACTORS R-101..R-104; THIS SHEET = R-101 (OTS UNIT).

2. RECIPE (ASSUMED OTS DESIGN POINT): 60 t DEMIN WATER 55 C + PVA (HV-107), 51 t VCM, INITIATORS A + B (11/2 1.5 h / 12 h AT 57 C), 57.0 C (GRADE K 67). END POINT = PRESSURE DROP 1.8 bar -> 85 % CONVERSION, 43.2 t PVC, REACTION 5.3 h; CYCLE -7 h -> 4 x 43.2 t x 8000 / 6.9 h = 200 kJ/y.

3. VESSEL D 4.5 m x T/T 7.4 m (141.5 m3), SS-CLAD. SHELL 15.2 barg & FULL VACUUM / 93 C, JACKET 5.2 barg (220 / 75 psi, 200 F). PSV-100 SET 15.0 barg TO DUMP TANK D-103; RUNAWAY / TWO-PHASE RELIEF SIZING TBC (DIERS METHOD).

4. BATCH SEQUENCE (DCS): XV-109 EVACUATE TO <= 0.05 bar a -> XV-101 WATER (FQ PRESET) -> XV-102 VCM (PERMISSIVES: XV-106 SHUT, EVACUATED, WATER >= 95 %, M-101 RUNNING, FI-110 >= 300 m3/h) -> HEAT-UP -> XV-103 INITIATOR AT TI-100 >= 54 C -> REACTION -> END POINT -> XV-104 SHORT-STOP -> XV-105 / PV-105 DEGAS BELOW 2 barg -> HV-141 N2 PAD (1.5 barg) -> XV-106 DISCHARGE. ALL XV FAIL CLOSED; SEQUENCE SIGNALS NOT SHOWN.

5. E-101: VCM VAPOUR UP AND CONDENSATE BACK DOWN THE SAME 18" NOZZLE. INERTS (N2 / AIR) COLLECT AT THE TOP AND BLANKET THE TUBES (TI-132 FALLS TOWARD CW TEMPERATURE) - VENT WITH HV-135. E-101 CW BY VPC-130 / FIC-130 ON PVC-PID-002 (ANTI-FOAM LIMIT 12 m3/h PER % XI-100).

6. XI-100 = CALORIMETRIC CONVERSION IN DCS = INTEGRAL(JACKET + BAFFLE + E-101 DUTIES - AGITATOR POWER + HEAT STORED) / (dHp x FQ-102).

7. LIC-100 (RADAR) HOLDS THE LEVEL WITH SHRINKAGE-COMPENSATION WATER (FIC-103 IN CASCADE): PVC 1.40 vs VCM 0.85 t/m3 -> ABOUT 20 m3 INJECTED PER BATCH; KEEPS THE JACKET WETTED.

8. SIS I-1..I-6 IN A SEPARATE LOGIC SOLVER (TABLE). SHORT-STOP POT D-105 IS N2-PRESSURISED; SHORT-STOP ONLY MIXES WITH THE AGITATOR RUNNING (< 30 s) - AGITATOR ON EMERGENCY POWER (TBC).

9. LINE SIZES AND MATERIALS PRELIMINARY - TBC BY PIPING. SERVICE CODES: PW PROCESS WATER, VC VCM LIQUID, VG VCM GAS, IN INITIATOR, SS SHORT-STOP, SL SLURRY, NG NITROGEN, CW COOLING / JACKET WATER.

LEGEND

	PROCESS LINE		UTILITY LINE (CW, STEAM, N2)
	SIGNAL LINE		LINE JUMP (NO JOIN)
	CONTROL VALVE FO / FC		ON/OFF VALVE (SEQ. / SIS)
	MANUAL VALVE		ORIFICE FLOW ELEMENT
	DCS INSTRUMENT		FIELD INSTRUMENT
	SIS FUNCTION		DCS FUNCTION BLOCK
	OFF-SHEET LINK		RELIEF VALVE

CONTROL LOOP SCHEDULE

LOOP	MEASURES	MOVES	FAIL	SET / NOTE
FIC-101	WATER CHARGE	FV-101	FC	240 m3/h, FQ 60 t
FIC-102	VCM CHARGE	FV-102	FC	180 t/h, FQ 51 t
LIC-100	R-101 LEVEL	FIC-103 SP	-	83 %
FIC-103	INJECTION WATER	FV-103	FC	CAS (~3 m3/h)
TIC-100	R-101 TEMP.	TIC-110 SP	-	57.0 C (K 67)
TIC-110	JACKET INLET	TY-110	-	CAS (~38 C)
TY-110	0-50 % OUT	TV-110A STEAM	FC	HEAT-UP
TY-110	50-100 % OUT	TV-110B CW	FO	COOLING
FIC-120	BAFFLE CW	FV-120	FO	80 m3/h
VPC-130	TIC-110 OUT	FIC-130 SP	-	50 % (NOTE 5)
FIC-130	E-101 CW	FV-130	FO	CAS
HIC-135	-	HV-135 INERTS	FC	MANUAL
HIC-105	-	PV-105 DEGAS	FC	END OF BATCH

SIS INTERLOCKS

I-	INITIATOR	SETTING	ACTION
1	TSHH-101 R-101 TEMP.	> 63 C, 10 s	A
2	PSHH-101 R-101 PRESS.	> 12.5 barg, 5 s	A
3	XS-101 AGITATOR STOP	REACTING, 15 s	A
4	FSSL-110 JACKET FLOW	< 120 m3/h, 30 s	A
5	LSHH-100 R-101 LEVEL	> 94 %, 5 s	B
6	HS-001 ESD PUSHBUTTON	MANUAL	A + XV-108

A = SHORT-STOP XV-104, CLOSE XV-101/102/103, FV-103 TO 0, FORCE TV-110B / FV-120 / FV-130 OPEN. B = CLOSE XV-101/102, FV-103.

ITEM (OTS DESIGN POINT)	VALUE
WATER / VCM CHARGE	60 t (55 C) / 51 t (25 C)
REACTION	57.0 C, 8.6 barg
LEVEL / CONTENTS AT 2 h	83 % / 117.8 t
Q REACTION AT 2 h	2.71 MW (PEAK 3.43)
JACKET / BAFFLES / E-101	1.48 / 0.40 / 0.98 MW
END POINT	dP 1.8 bar, X 0.85
PVC PER BATCH / TIME	43.2 t / 5.3 h

REV	DESCRIPTION	DATE
A	ISSUED FOR OTS DEVELOPMENT (REVIEW)	25-SEP-26

CHULA CHEM. ENG. - PVC REACTOR VER 1 (OTS)

P&ID - R-101 PVC REACTOR, CHARGES, CONDENSER, VENTS

DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --	DATE: 25-SEP-26
DWG NO. PVC-PID-001	REV A	SCALE N.T.S.	SHEET 1 OF 2